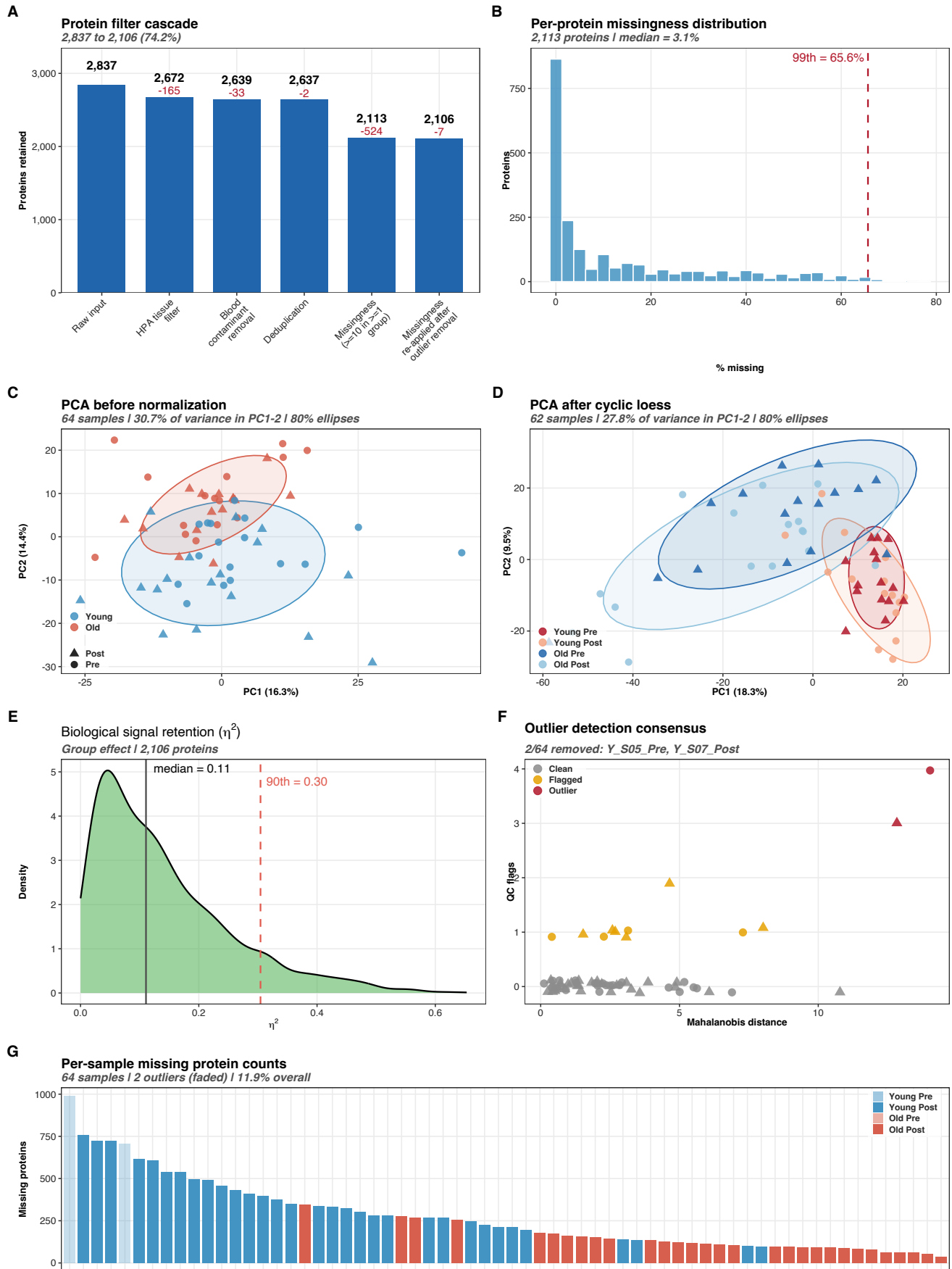


## Pipeline QC — Pre-Processing

DIA-NN → HPA filter → cyclic loess → outlier removal | 2,837 → 2,106 proteins × 62 samples



S1a Figure. Quality-control and normalization pipeline. (A) Protein filter cascade (2,837 to 2,106). (B) Per-protein missingness. (C) Pre-normalization PCA. (D) Post-normalization PCA. (E) Eta-squared variance decomposition. (F) Outlier consensus scoring. (G) Per-sample missingness.